

Bayesian decision theory

- Problem definition
- Point-estimation problems

BO

In BO, we rely on probabilistic surrogates of the objective function

We evaluate the objective only at **query points** and use them to

- refine the surrogate
- approximate the optimum and the optimizer
- decide on termination

We use the surrogate to guide the acquisition of **subsequent queries**

Termination and query selection are actions we take in view of the current surrogate which depends upon our past queries and, in this way, our past actions

Because our objective function is unknown, both actions give rise to **(sequential) decision problems**

In BO, the decision problems are **inherently Bayesian**

Decision problems

A **decision problem** involves three spaces

- space of **states** $\mathcal{Q}$...this is our state-space... ...our states live here, $q \in \mathcal{Q}$
- space of **data** $\mathcal{W}$...this is our data-space... ...our data live here, $w \in \mathcal{W}$
- space of **actions** $\mathcal{A}$...this is our action-space... ...our actions live here, $\alpha \in \mathcal{A}$

and asks for

- a **decision rule**

$$D : \mathcal{W} \mapsto \mathcal{A}$$

that selects an action $D(w) = \alpha \in \mathcal{A}$ given data $w \in \mathcal{W}$

Decision rules are derived through **loss functions**

$$L : \mathcal{A} \times \mathcal{Q} \mapsto (-\infty, +\infty)$$

The loss $\ell = L(\alpha, q)$ encodes the consequences of action α provided our state attains value q

...higher losses mean worse outcomes
...lower losses mean better outcomes

Bayesian decision problems

A Bayesian decision problem consists of

- a **Bayesian model** that couples states and data

$$q \sim Q$$

$$w|q \sim P_q$$

...our model's prior Q is a probability distribution on the state-space $\mathcal{Q}$

...our model's likelihood $\{P_q\}_{q \in \mathcal{Q}}$ is a family of probability distributions on the data-space $\mathcal{W}$

- a **loss function** that couples states with actions

$$L : \mathcal{A} \times \mathcal{Q} \mapsto (-\infty, +\infty)$$

...our model's loss L is a function on the joint action/state-space $\mathcal{A} \times \mathcal{Q}$

Our **main goal** is to derive a decision rule

Bayesian decision problems

Given a Bayesian decision problem

$$q \sim Q$$
$$w|q \sim P_q$$

The **posterior expected loss** of action α under data w is given by

$$\rho(\alpha, w) = \underbrace{\mathbb{E}_{q|w} L(\alpha, q)}_{\text{posterior expectation}}$$

$$\rho(\alpha, w) = \underbrace{\int_{\mathcal{Q}} dq L(\alpha, q) p(q|w)}_{\text{continuous states}}$$

$$\rho(\alpha, w) = \underbrace{\sum_{q \in \mathcal{Q}} L(\alpha, q) p(q|w)}_{\text{discrete states}}$$

A **Bayes action** α_w^* under data w is the minimizer

$$\alpha_w^* = \operatorname{argmin}_{\alpha \in \mathcal{A}} \rho(\alpha, w)$$

and the **Bayes rule** is

...this decision rule selects **only** Bayes actions

$$D^*(w) = \alpha_w^*$$

Point estimation

In point-estimation, we seek to estimate a single value $\hat{\theta}$ for an unknown parameter θ of interest

- First, we develop a Bayesian model

$$\begin{aligned}\theta &\sim \Pi \\ \omega_n|\theta &\sim \Lambda_\theta, & n = 1, \dots, N\end{aligned}$$

for appropriate prior Π and likelihood $\{\Lambda_\theta\}_\theta$

in this problem, the state-space is the parameter-space and the data-space is composite

- Then, we acquire our data $\omega_{1:N}$
- Finally, we choose a value $\hat{\theta}(\omega_{1:N})$ from the parameter space in response to our acquired data $\omega_{1:N}$

in this problem, the action-space is the parameter-space

In the last step, we need to solve a decision problem

...which is Bayesian

Our decision problem is completed with a loss function and the derivation of the associated Bayes estimator which minimizes the posterior expected loss

Point estimation

We seek a point-estimate for a **continuous scalar parameter** of interest $\theta \in \mathbb{R}$ under the **squared deviation loss**

$$L(\hat{\theta}, \theta) = (\hat{\theta} - \theta)^2$$

Our **posterior expected loss** is given by

$$\begin{aligned}\rho(\hat{\theta}, \omega_{1:N}) &= \mathbb{E}_{\theta|\omega_{1:N}} L(\hat{\theta}, \theta) \\&= \int d\theta L(\hat{\theta}, \theta) p(\theta|\omega_{1:N}) \\&= \int d\theta (\hat{\theta} - \theta)^2 p(\theta|\omega_{1:N}) \\&= \int d\theta (\hat{\theta}^2 - 2\hat{\theta}\theta + \theta^2) p(\theta|\omega_{1:N}) \\&= \hat{\theta}^2 \int d\theta p(\theta|\omega_{1:N}) - 2\hat{\theta} \int d\theta \theta p(\theta|\omega_{1:N}) + \int d\theta \theta^2 p(\theta|\omega_{1:N}) \\&= \hat{\theta}^2 - 2\hat{\theta} \underbrace{\mathbb{E}_{\theta|\omega_{1:N}} \theta}_{\text{posterior mean}} + \int d\theta \theta^2 p(\theta|\omega_{1:N})\end{aligned}$$

Point estimation

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Our **posterior expected loss** is given by

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The minimizer of the posterior expected loss solves

$$\begin{aligned} 0 &= \frac{\partial}{\partial \hat{\theta}} \rho(\hat{\theta}, \omega_{1:N}) \\ &= \frac{\partial}{\partial \hat{\theta}} \left(\hat{\theta}^2 - 2\hat{\theta} \mathbb{E}_{\theta|\omega_{1:N}} \theta + \int d\theta \theta^2 p(\theta|\omega_{1:N}) \right) \\ &= \left(\frac{\partial}{\partial \hat{\theta}} \hat{\theta}^2 \right) - 2 \left(\frac{\partial}{\partial \hat{\theta}} \hat{\theta} \right) \mathbb{E}_{\theta|\omega_{1:N}} \theta + \left(\frac{\partial}{\partial \hat{\theta}} \int d\theta \theta^2 p(\theta|\omega_{1:N}) \right) \\ &= 2\hat{\theta} - 2 \mathbb{E}_{\theta|\omega_{1:N}} \theta \end{aligned}$$

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$$\rho(\hat{\theta}, \omega_{1:N}) = \hat{\theta}^2 - 2\hat{\theta} \mathbb{E}_{\theta|\omega_{1:N}} \theta + \int d\theta \theta^2 p(\theta|\omega_{1:N})$$

Our **Bayes estimator** is

$$\hat{\theta} = \mathbb{E}_{\theta|\omega_{1:N}} \theta$$

Summary of point estimation

We seek a point-estimate for a **continuous scalar parameter** of interest $\theta \in \mathbb{R}$

with the **squared deviation loss**

$$L(\hat{\theta}, \theta) = (\hat{\theta} - \theta)^2$$

the **Bayes estimator** is

$$\hat{\theta} = \begin{pmatrix} \text{posterior} \\ \text{mean} \\ \text{of } \theta \end{pmatrix}$$

with the **absolute deviation loss**

$$L(\hat{\theta}, \theta) = |\hat{\theta} - \theta|$$

the **Bayes estimator** is

$$\hat{\theta} = \begin{pmatrix} \text{posterior} \\ \text{median} \\ \text{of } \theta \end{pmatrix}$$

with the **relaxed 0-1 loss**

$$L_{\epsilon}(\hat{\theta}, \theta) = \begin{cases} 1, & |\hat{\theta} - \theta| \geq \epsilon \\ 0, & |\hat{\theta} - \theta| < \epsilon \end{cases}$$

the (limiting) **Bayes estimator** is

$$\lim_{\epsilon \rightarrow 0} \hat{\theta}_{\epsilon} = \begin{pmatrix} \text{posterior} \\ \text{mode} \\ \text{of } \theta \end{pmatrix}$$